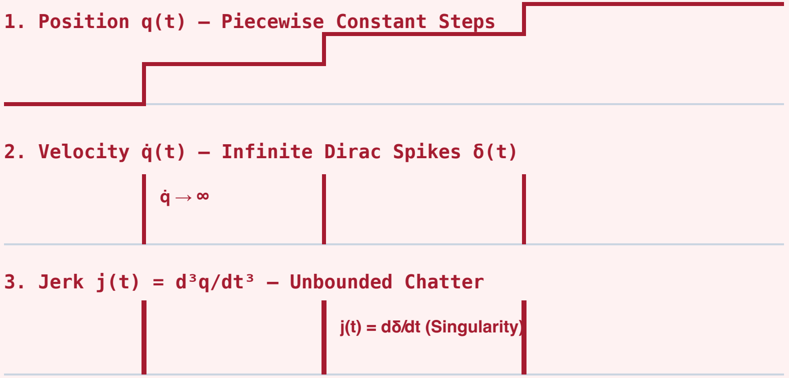
DERIVATIVE CONTINUITY (C^2) VS. DISCONTINUOUS WAYPOINT STEPPING

How quintic polynomial spline interpolation bounds acceleration and eliminates infinite jerk spikes (δ(t))

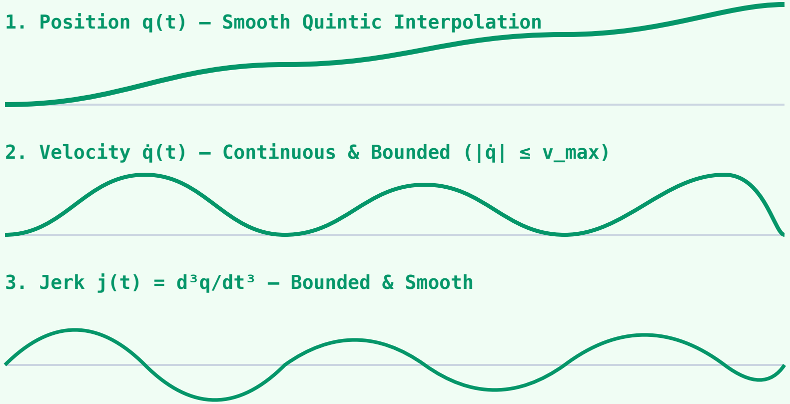
❌ DISCONTINUOUS C0 STEPPING (ZERO-ORDER HOLD)



PHYSICAL HARDWARE CONSEQUENCES:

- **Extreme Torque Spikes:** Motor commanded to accelerate instantly.
- **Current Inrush ($I^2 R$):** Massive coil heating, blowing power fuses.
- **Gearbox Backlash & Wear:** Strips reduction gear teeth.
- **Audio Acoustics:** Loud, high-pitched mechanical knocking.

✓ C2 CONTINUITY (QUINTIC POLYNOMIAL SPLINE)



PHYSICAL SYSTEMS ADVANTAGES:

- **Smooth Motor Commutation:** Current follows gradual sinusoidal curves.
- **Thermal Protection:** Operates within nominal continuous torque ratings.
- **10x Extended Lifespan:** Zero harmonic resonance or gearbox damage.
- **Rock-Solid Tracking:** 1 kHz MCU PID loops track trajectory perfectly.